

An interface approach to island constraints¹

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Abstract: Island constraints are one of the central topics in the study of generative grammar. Since their discovery by Ross (1967), various hypotheses have been raised to uncover the nature of these constraints. However, many issues remain unresolved. This paper reevaluates the properties of subject islands, complex noun islands, and adjunct islands within the framework of the labeling theory. Based on Chomsky (2015) and Bošković (2016), it proposes that islands do not originate from syntactic restrictions but rather from interpretive failures. This paper not only provides a fresh perspective for island research but also offers a unified account of strong islands without disregarding the semantic factors involved in adjunct island, making a novel contribution to the field. Furthermore, it successfully accounts for some exceptional extractions out of islands and their cross-linguistic differences.

Key words: Islands, interface, labeling, weak islands

1. Goals

- (i) Use the labeling theory to uniformly account for strong islands, offering a fresh perspective on the nature of island constraints.
- (ii) Explain exceptional extractions out of islands, as well as their cross-linguistic differences.

2. Background

There are some contexts from which a constituent cannot be extracted, as shown in (1), and these contexts are called “islands”.

- (1) a. *Which boy’s guardian’s did we elect employer president?
- b. *Which boy’s did we elect guardian’s employer president?
- c. *Which exams did John date a lot of girls [instead of doing]? (Cattell 1976: 38)

Numerous hypotheses have been proposed to explain them, including the Subjacency Constraint (Chomsky 1977), the Condition on Extraction Domain (Huang 1982), the Empty Category Principle (Kayne 1981), and Barriers (Chomsky 1986). These hypotheses rely on terms such as ‘government’ that lack conceptual necessity.

In the era of the Minimalist Program, many efforts have been made to attribute islands to constraints on various syntactic operations, including the Merge operation (Sabel 2002), the Agreement operation (Rackowski & Richards 2005), the spell-out operation (Uriagereka 1999, 2011; Fox & Pesetsky 2005), late insertion (Stepanov 2007), or constraints on syntactic derivations, such as the labeling approach to antilocality (Bošković 2015, 2016, 2017), Chain Uniformity (Takahashi 1994, Richards 2001), the Phase Impenetrability Condition (Chomsky 2008, Müller 2011), and Relativized Minimality (Rizzi 1990, 2011). While these proposals are indeed ingenious, they still exhibit notable weaknesses. Empirically, all these proposals predict

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that subject islands, complex NP islands, and adjunct islands are opaque to extraction. However, this is not always true because exceptional extraction out of islands can be noted in almost every language, as demonstrated by the examples below.

(2) a. This is a paper that we really need to find someone who understands. (Chung & McCloskey 1983: 708)

b. John is the sort of guy_i that I don't know a lot of people who think well of t_i. (Culicover 1999: 230)

There are also researches that focus on specific islands. For instance, Brown (2017) examines the adjunct island, arguing that only adjuncts merged in a position lower than v/v* are porous because they can be c-commanded by the head v and establish agreement. Nevertheless, this cannot explain the facts shown below.

(3) a: Which book did John design his garden [after reading]?

b: An introduction to landscape gardening.

(4) a: Which book did John design his garden [after reading]?

b: #Finnegans Wake. [purposive adjuncts]

3. A specific research: Bošković (2016)

Bošković (2016) provides a uniform account of island constraints using Chomsky's (2013, 2015) labeling theory. His main hypotheses are as follows.

(5) Timing of labeling:

(a) When a head and a phrase are merged, the head provides a label for the derived structure immediately.

(b) When two phrases are merged, the derived structure will not be labeled until the phrase is sent to the interface.

(6) Antilocality: Movement of A targeting B must cross a projection distinct from B (where unlabeled projections are not distinct from labeled projections).

With these assumptions, he succeeds in accounting for strong islands. Take (7) as an example. If *who* is to move out of the subject, it must first move to the edge of the subject, which makes the subject phrase and the TP unlabeled before they are spelled out, as they consist of two phrases. The unlabeled structure is labeled '?', as in (8).

(7) *I wonder who_i [friends of t_i] left.

(8) [? [? who [DP friends of t_i]] [T T [vP left]]]

When *who* moves to merge with the targets C, it only crosses the labeled node C, as shown in (9), thereby violating the antilocality condition.

(9) Who_i [C C [? [? t_i [DP friends of t_i]] [TP T [vP left]]]

Problems: first, in accordance with his analysis, if a sentence violates the antilocality condition, it will be unacceptable, as in (10)a and (11)a. However, once the structure containing the complex NP island is elided, such movement becomes acceptable, as demonstrated in (10)b and (11)b.

(10) a. *I don't remember which (Balkan language) they want to hire some who speaks.

b. They want to hire someone who speaks a Balkan language, but I don't remember which. (Merchant 2001: 87)

(11) a. *Ben will be mad if Abby talks to one of the teachers, but she couldn't remember which (of the teachers) Ben will be mad if she talks to.

b. Ben will be mad if Abby talks to one of the teachers, but she couldn't remember which. (Merchant 2008:136)

A lot of evidence, including preposition stranding and Case matching, suggests that the remnants in ellipsis constructions are derived by movement (see Merchant 2001, among others). Since the antilocality condition is a syntactic constraint, any movement that violates this constraint should be prohibited. Consequently, *which* in (10) should not be able to move out of the complex NP island licitly. In other words, if (10) is ungrammatical because it violates a syntactic constraint, it should not be salvaged by ellipsis (see Ernst 2022 for a similar argument). Second, this analysis cannot account for the issue related to exceptional extraction out of an adjunct. For example, movement of *what* from the adjunct is allowed in (2) and (12).

(12) What did you go there to discuss___?

4. Our hypotheses

First, an extension of Bošković's (2016) timing of labeling is proposed: when two constituents are merged, the derived structure will be labeled as early as possible. This principle may be called the Earliness Principle of labeling (a reflection of the economy of syntactic derivation) (Bošković 2016). The Earliness Principle of labeling can be manifested in the following two ways:

(i) When a strong head merges with a phrase, it will label immediately. This is consistent with Bošković (2016).

(ii) When two phrases are merged, once a situation similar to (i) arises, the labeling algorithm can be applied before a structure is spelt out. Specifically, suppose that two phrases are merged. If one phrase cannot label temporarily during the derivational process while the other can, and the head of the phrase that can label agrees (or has a feature-sharing relationship) with the other, this situation mirrors the scenario where a head merges with a phrase: the head can label, but the phrase cannot; the head c-selects the phrase (Bošković 2016, Neeleman et al. 2023). Due to this similarity, once such a structure is formed, the labeling algorithm can be applied immediately.

For illustration, consider the merger of XP and YP. Once the phrase AP moves to the edge of XP, the derived structure is temporarily unlabeled because it comprises two phrases, as in (13). If the head Y agrees with or has a feature-sharing relationship with XP, then the situation is similar to the merger of the head H and the phrase QP shown in (14). In this case, Y will provide a label for structure α immediately (see (13) and (15)).

(13) [α [$_{\text{P}}$ AP [$_{\text{XP}} \dots X \dots t_i$] [$_{\text{YP}} \dots Y \dots$]]

(14) [$_{\text{H}}$ H QP]

(15)

The first tree diagram shows a head 'H' at the top, with two lines branching down to 'H' and 'QP'. The second tree diagram shows a head 'Y' at the top, with two lines branching down to '?2' and '...Y ...'.

Second, a new definition of antilocality is needed, as shown in (16). This condition follows from the third factor constraint, as Hornstein (2024) points out that distance is computed based on syntactic nodes.

(16) Distinctness condition: a constituent cannot cross two identical nodes successively.

Based on the assumptions above, we argue that island constraints arise because, to avoid violating the syntactic constraint, a head is forced to provide a label for its mother node with the help of the Earliness Principle, regardless of whether the label it provides is appropriate. If a head is unable to provide a label for its mother node, the label it offers will not be conducive to

interpretation at the interface. As a result, interpretation problems will arise. In other words, the island constraint does not result from syntactic constraints but from interpretation problems at the interface.

5. The account of island constraints

5.1 The Subject islands

(17) is derived as (18) shows. After *who* moves to Spec-CP, the phrase in the subject position cannot be labeled temporarily and is marked as ‘?1’ for convenience. The complement of C, namely the traditional TP, cannot be labeled as it consists of two phrases and is marked as ‘?2’.

(17) *Who_i did that Mary kissed t_i bother you?

(18) [C [_{?2} [_{?1} who_i [CP that Mary kissed t_i] [_α T bother you]]]]

If *who* moves from the embedded Spec-CP, it will move across two unlabeled nodes consecutively, and then the distinctness condition will be violated. However, there is a way to evade this constraint. Consider (18) again. Since T agrees with the subject, due to the Earliness Principle of labeling, T will try to provide a label for the structure labeled as ‘?2’ before the Transfer operation takes effect. Consequently, the structure in (18) should be labeled as shown in (19) shows. Then, the movement of *who* will not violate the syntactic constraint.

(19) [C [_T [_{?2} who_i [CP that Mary kissed t_i] [_α T bother you]]]]

However, T is not a strong head in English, and the label provided by T cannot be properly interpreted at the interface, leading to the low acceptability of (17).

Consider the island repair phenomena under ellipsis. The complete form of (20) should be (21). In (20)/(21), *which one* can move out of the subject island without violating syntactic constraints. Nevertheless, such a movement forces T to provide an improper label. If the structure in the bracket is elided before Spell-out (Baltin 2012), there will be no improper labels at the interface. Therefore, no problem will arise in (20).

(20) That certain countries would vote against the resolution has been widely reported, but I’m not sure which ones.

(21) *That certain countries would vote against the resolution has been widely reported, but I’m not sure which ones [that t would vote against the resolution has been widely reported]. (Chuang et al 1995: 272)

Consider the exceptional extraction out of subject islands, as exemplified by (22)).

(22) Mary [a kltukl [el kmo ng-oltoir er a John ___i]] (Palauan)

Mary R-clear comp R-3sg-imp-love John

“Mary, [that ___ loves John] is clear. (Georgopoulos 1991)

Georgopoulos (1991), Lasnik & Saito (1992), Yatsushiro (1999), and Stepanov (2007), among others, have all pointed out that in these languages, sentential subjects can remain within vP. Assuming their analysis is reasonable, the process whereby an element moves out of a sentential subject can be briefly outlined as shown in (23).

(23) [_v [_{? XP_i [CP ... t_i...]]] [_v v ...]}

Since v can have an agreement relationship with the phrase in its Spec in languages where the subject stays in situ, v should be able to provide a label for the structure shown in (23), and the movement of XP to the matrix clause will not violate the distinctness condition.

Based on the argument above, we infer that if T is a strong head, the extraction of a phrase from Spec-TP should be allowed because such extraction does not violate the distinctness

condition, as demonstrated in (24).

(24) $[_T [_? XP_i [_{CP} \dots t_i \dots]] [_{T'} T]]$

The above prediction is confirmed by facts in East Asian languages like Chinese and Korean, as shown in (25). In these languages, pro-drop is allowed, indicating that T should be a strong head (see also Tian et al. 2024).

(25) Zhexie shi_i wo juede ta shuo t_i bu heshi.

these thing I think he say not proper

‘These things, I think it is not proper for him to say.’ (Xu and Langendoen 1985: 13)

Sentences (26) and (27) illustrate that adjacency between C and T permits the extraction of a phrase from the subject. In these scenarios, C and T are treated as a single constituent, with C functioning as a copy of T (Gallego 2014). T is characterized by an EPP/edge feature and an uninterpretable phi-feature, which designate it as a weak head. However, when T and C are separated by intervening heads, as depicted in (28), the composite head <C, T> cannot be formed (Park 2024). In such cases, T is only endowed with the EPP/edge feature, lacking inherent uninterpretable phi-features. Before T acquires uninterpretable features from C, it operates as a strong head. Based on the Early Labeling Principle, T can serve as a licit head prior to the structure being spelled out in (27).

(26) *[Of which car]₁ did [the driver ____₁] cause a scandal?

(27) (?)Of which car, according to your recollection, did the driver cause a scandal?

(28) $[_{FocP} PP_{[wh]}{}_1 Foc [_{XP} X [_{TP} [_{DP} \dots \text{---}_1 \dots] T \dots]]]$ (Zyman 2021:19)

5.2 The Complex NP islands

Consider (29). It should be derived as follows. First, *what* moves to Spec-CP, rendering the entire structure temporarily unlabeled, as shown in (30).

(29) ?What_i did you hear rumors that John bought t_i?

(30) $[_? \text{what} [_{CP} \text{that John bought } t_i]]?$

Subsequently, the entire structure merges with the nominal root RUMOR. Considering that NP is a phase (Bošković 2014, 2016, 2017, Sichel 2018), *what* should move to the edge of this phase. To enable this movement to circumvent the distinctness condition, the root RUMOR should try to provide a label for its mother node the moment it merges with CP, as (31) shows. Since the lexical root is too weak to label, the label provided by RUMOR cannot be interpreted at the interface, resulting in unacceptability.

(31) $[_{What_i} [_{RUMOR_{RUMOR}} [_? t_i \text{that} [_{TP} \text{John bought } t_i]]]]?$

The lexical head V should also be weak. However, a constituent can move across the edge of such a head, as shown in (32). Why?

(32) What do you think that he will buy?

As argued by Epstein et al. (2016), when a verb selects a clause as its complement, it first pair-merges with v. The derived complex head then merges with a clause. Building on Epstein et al. (2016), I assume that the embedded clause in (32) merges with the amalgam v-THINK, as shown in (33). When *what* moves to the edge of v-THINK, the distinctness condition is not violated.

(33) $\text{what} [_{v\text{-THINK } V\text{-THINK}} [_{??} t_{\text{what}} [_{CP} \text{that he will v-buy } [_{\text{phi-phi}} [t_{\text{what}} \text{buy}]]]]]]?^2$

² Based on Baker (1988) and Radford (1997), I place the theme argument in Spec-VP. If the theme argument were placed in the complement of V, its movement to Spec-VP would violate the antilocality condition, and extraction of a constituent from the theme argument would violate the freezing condition, contrary to facts.

Our assumption can also explain the contrast shown below.

(34) Who_i did they see (some) friends of t_i yesterday?

(35) ?*Who_i did there arrive (some) friends of t_i last week?

These sentences differ in the properties of the verbs: the first one is transitive and the second one is unaccusative. The first one is a phase head, while the second one is not (Chomsky 2001, 2008). This might be at the core of the problems. The derivational process of these two sentences can be demonstrated with (36) and (37). In (34)/(36), the lower copy of SEE can inherit phasehood. When *who* moves to the edge to VP, the feature matching between lexical verb and the phrase *some friends of t_i* can have V strengthened, allowing it to project a licit label (see Chomsky 2015). In contrast, the lower copy of ARRIVE cannot inherit phasehood, and *what* will move across two successive unlabeled nodes, violating the distinctness condition.

(36) v-see [_V [_? what [(some) friends of t_i] [SEE yesterday]]?

(37) ?*Who_i [C [_T T [v- ARRIVE [_? [_? who (some) friends of t_i ARRIVE] last week?

A speculative account can be provided for the contrast shown in (38)a and (38)b.

(38) a. *Who did Kerry hear the rumor that Kelsey is fond of?

b. Who did Kerry start the rumor that Kelsey is fond of? (Davies & Dubinsky 2002: 31-32)

Based on Baker (1988) and Davies & Dubinsky (2002), it can be argued that the head of the NP (i.e., the root RUMOR) in (38)b undergoes noun incorporation with v*, which renders the definite effect of the nominal phrase disappear. In the phonological system, the lower copy of the movement head gets pronounced (Hornstein 2024). After the incorporation, the c-selection feature matching between v* and RUMOR making the latter strong. At the interface, the Form Copy operation establishes a copy relationship between the copies of RUMOR (see Chomsky 2021, 2023). Accordingly, the lower copy of RUMOR can be interpreted.

5.3 The adjunct island

Let us see (39). Since CP is a phase, *what* should first move to the edge of CP, yielding an unlabeled structure marked as “?1”. The whole structure then merges with vP *was angry*, forming another unlabeled structure, marked as “?2”, as shown in (40).

(39) *What was she angry [after he bought]? (Rackowski & Richards 2005: 595)

(40) [_? [_{VP} was angry] [_{?1} t_{what} [_{CP} after he bought t_{what}]]]?

If *what* moves to the Spec position of the matrix clause, it will move across two unlabeled nodes consecutively, which is marked with a box in (41). Consequently, the distinct condition is violated.

(41) [what [[_{?2} [_{VP} was angry] [_{?1} t_{what}]] [_{CP} after he bought t_{what}]]]?

To avoid violating the distinctness condition, the earliness principle of labeling will come into effect. Suppose the derivation reaches the stage shown in (40). With the aid of the earliness principle of labeling, v is chosen as the label for the structure marked as “?2.” The movement of *what* satisfies the distinctness condition, as shown in (42).

(42) [what [_v [_{VP} was angry] [_? t_{what} [_{CP} after he bought t_{what}]]]]?

Why is (39) unacceptable? The issue arises because, although we take the head v as the label for the structure formed by vP and the adjunct, we neglect a crucial aspect of labeling economy. As mentioned early, if a head provides a label for its mother node based on the earliness principle of labeling, it should bear certain selection/agreement relationship with the other projection. In (39), although v is a strong head, it does not have any feature selection/agreement relationship with the adjunct. Therefore, it is not a competent label. Specifically, once v is considered the label

for the structure in (40), at the interface, the structure will be interpreted as if that *v* had certain feature selection/ agreement relationship with the adjunct, which is not the case.

Exceptional extraction of a constituent, as shown in (43), falls right into place.

(43) What did Mary lie about singing what all day?

The derivational process of this sentence is illustrated in (44).

(44) what [_v *v* [go there] [?? *t*_{what} [_{CP} to discuss *t*_{what}]]]?

Although there is no morphological agreement with the *vP/v* and its adjunct, there exists a special relationship between them, which is formulated as the single event constraint shown in (45).

(45) The Single Event Grouping Condition: An instance of *wh*-movement is legitimate only if the minimal constituent containing the head and the foot of the chain can be construed as describing a single *event grouping*. (Truswell 2011:157).

In accordance with Den Dikken (2018: 126-130), when a matrix clause and an adjunct can form a single event, the light verb *v* and the adjunct agree in Aspect (see also Browning 1987, Uriagereka 1988, Boeckx 2003, Ernst 2022). Such an agreement ensures that *v* can utilize the Earliness Principle to provide a label for the structure formed by *vP* and the adjunct.

6. More on exceptional extraction out of islands

Consider the additional exceptional extraction from the Complex NP Island, as shown in (46).

(46) This is a paper that we really need to find someone who understands. (Chung & McCloskey 1983: 708)

As for these facts, Sichel (2018) argues that the relative clause (RC, henceforth) can be classified into two types: raising RC and matching RC, which are structured as follows.

(47) a. [_{DP} *D* [_{NP₁} [_{CP} ~~*NP₁*~~ [... *T*_{NP}...]]]] (matching RC)

b. [_{DP} *D* [_{CP} ~~*NP₁*~~ [... *T*_{NP}...]]] (raising RC)

If a constituent is to move out of a matching RC, it will first adjoin to the NP (i.e., the higher *NP₁* in (47)a) because NP is a phase. Movement from NP to *D* violates the antilocality condition. However, movement of a constituent from an in-situ raising RC does not violate the antilocality condition because the whole CP is crossed.

Following Preminger (2010), she also assumes an articulated CP hypothesis, in which CP dominates another A'-projection, such as FocP. Simply put, if a DP is to be extracted from a RC, its movement can be outlined as shown in (48).

(48) [_{CP₂} *DP₂* ... *V* [_{DP} ~~*DP₂*~~ [_{D⁰} [_{CP₁} ~~*DP₂*~~ [_{FocP} *DP₁*... [_{IP} ~~*DP₄*~~...*DP₂*...]]]]]]]

First, *DP₁*, the head of the RC, moves to Spec-FocP, and then *DP₂* moves to Spec-*CP₁*, from which it moves to the edge of the phase DP without violating the antilocality constraint.

Sichel's (2018) account relies on illicit movement from the edge of NP to that of DP. However, in languages without articles, such as Serbo-Croatian, the Complex NP Island effect also exists, as shown in (49). In these languages, there is no DP (Bošković 2012).

(49) ?? Štai si čuo [_{NP} glasinie [_{CP} da je Ivan kupio *t_i*]]?

what are heard rumors that is Ivan Bought

'What did you hear rumors that Ivan bought?' (Bošković 2015: 607)

These facts can be explained within our framework. First, consider the extraction of a constituent from a matching RC or nominal complement clause, as shown in (49). As explained in section 5.2, part of this process can be outlined as follows. To prevent the movement of Štai 'what' from violating the distinctness condition, the root RUMOR must project. However, being a

weak head, its label cannot be used for interpretation, leading to interpretative failure.

(50) [What_i [RUMOR_{RUMOR} [? t_i that [TP Ivan bought t_i]]]]?

Now consider the extraction of a constituent from a raising RC. According to Sichel (2018), the movement of the phrase *a paper* should be as (51) shows. Such a movement does not violate the distinctness condition, and there are no improper labels in the structure.

(51) a paper [DP D [? t_a paper [CP someone [FocP who [... T_{who-some}...t_a paper]]]]

(52) is unacceptable because the object is a presupposed constituent. Such a constituent should undergo cover movement (see Sichel 2018). Even if the object-modifying RC can be analyzed as a raising RC, the movement of *what* out of the RC violates the freezing condition.

(52) *What do you like the person that John met t_i?

7. Other phenomena

Sentences like (53) seem to pose a challenge to our distinctness condition. In (53) there are two adjacent adjuncts. It is assumed above that when two phrases are merged, their mother node is not labeled until the structure is spelt out. Along this vein, the derivation of (53) appears to be (54), where the movement of *what* violates the distinct condition.

(53) What did John hide upon receiving the warning yesterday?

(54) What_i did John [? [v_P hide t_i][upon receiving the warning] yesterday]]?

Instead of posing a challenge, this can support our hypothesis. Cinque (1999: 106) argues that adjuncts should be merged in the Spec position of functional heads. The following is a typical example, where Asp is split into several functional heads to accommodate adverbs.

(55) [no longer Asp_{terminative} [still Asp_{continuative} [just Asp_{retrospective} [soon Asp_{proximative} [briely Asp_{durative} [almost Asp_{prospective} [completely Asp_{SgCompletive(I)}]]]]]]]]

Following Cinque (1999), we can argue that v is split into several functional heads to accommodate the adjuncts, as (56) shows. As can be seen, the distinctness condition is observed.

(56) What_i did John [? [v₁ v₁ [? [v₂ v₂ [v_P hide t_i][upon receiving the warning] yesterday]]]]]]?

Our distinctness condition also entails that a head can have only one Spec (cf. Ura 1996). Facts like (57) can be explained without resorting to the multiple Specs hypothesis. We can adopt Nash & Rouveret's (1997) and Neeleman & Weerman's (1999) proxy category hypothesis by arguing that TP may be generated recursively, as (58) shows, with the higher TP considered a proxy category.

(57) John-ga musuko-ga (butai-de) odot-ta.

John-NOM son-NOM stage-at dance-PAST

'John's son danced at the stage.' (Ura 1994: 108)

(58) [TP Spec T [TP Spec T [v_P

To satisfy the distinctness condition, T in (58) should be strong. This is confirmed by the fact that in languages with multiple subjects, the subject can also be null. Our assumption that T in Chinese and Japanese is strong is compatible with Chomsky's (2015) understanding of weak heads. Essentially, Chomsky's (2015) hypothesis that English T is a weak head is a restatement of a very old view, which can be traced back to Taraldsen (1978) and Rizzi (1982). According to this perspective, languages allow subject drop because they have rich inflectional systems, while English does not allow subject drop due to its impoverished inflectional system. In other words, for T to be considered a weak head, it must have inflectional features (see also Huang 1982 and Saito 2007). Without Agr, non-finite Ts in English and T in Japanese and Chinese should be strong (Tian 2024).

In sum, our distinctness condition can capture the connection among Agr, pro-drop, and multiple subject constructions. It is also compatible with the cartographic theory, which can account for the linearization of natural languages.

8. Conclusion

In this paper, we reevaluate the nature of the complex NP island, the subject island, and the adjunct island within the framework of labeling theory. Based on Bošković (2016, 2017), we propose the distinctness condition, which requires that a constituent cannot move across two identical nodes successively. we also develop Bošković's (2016, 2017) timing of labeling by proposing an earliness principle in labeling: labels will be assigned as early as possible. With these principles, we demonstrate that complex NP islands, subject islands, and adjunct islands arise not because syntactic constraints are violated, but because driven by the distinctness condition, some heads that cannot label are forced to provide a label, leading to improper interpretation at the interface. In addition to offering a uniform account of island constraints, this approach can elucidate crosslinguistic differences in subject island extraction and exceptional extraction from complex NP islands and adjunct islands.

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